# Homework 9

1. Let $\{a_n\}_{n\geq 0}$ and $\{b_n\}_{n\geq 0}$ be two sequences of real numbers. Suppose that $0 < a_{n+1} \leq a_n$ for all $n \geq 0$ and $\sum_{n=0}^{\infty} b_n$ is convergent. Show that $\sum_{n=0}^{\infty} a_n b_n$ is convergent.

2. For which $x$ and $\theta$ does $\sum_{n=1}^{\infty} n^{-1} x^n \cos n\theta$ converge ?

3. For $n \in \mathbb{N}, x \in \mathbb{R}$ , put

$$f_n(x) = \frac{x}{1 + nx^2}$$

Show that $\{f_n\}$ uniformly converges to a function $f$, and that the equation

$$f'(x) = \lim_{n\to\infty} f_n'(x)$$

is correct if $x \neq 0$, but false if $x = 0$.

4. Construct sequences $\{f_n\}$, $\{g_n\}$ which uniformly converge on some set E, but such that $\{f_n g_n\}$ does not uniformly converge on E ( of course, $\{f_n g_n\}$ must converge on E).